DE–6397 **11**

DISTANCE EDUCATION

M.Sc. (Botany) (Specialization– Plant Biotechnology) DEGREE EXAMINATION, MAY 2013.

PLANT DIVERSITY

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

(5 × 20 = 100)

1. (a) Give an account of the economic importance of Rhodophyceae. (8)

   (b) Write a detailed account of reproduction in algae. (7)

   (c) Characterize the Cyanophyceae. (5)

2. (a) What are the various modes of nutrition in Fungi? (7)

   (b) Write about various types of spores in Fungi. (7)

   (c) Brief on the characteristics of deuteromycetes. (6)

3. (a) Give a short account on ecological importance of Lichens. (6)

   (b) Explain the structural variations in gametophytes of Bryophytes. (8)

   (c) Describe the classification of Bryophytes by Watson. (6)

4. (a) Elaborate the general characters of Pteridophytes.(8)

   (b) Explain the morphology of a Psylopsida member. (6)

   (c) Give a general account on origin Pteridophytes. (6)

5. (a) What are the general characters of Cycadales? (8)

   (b) Brief on the typical life cycle of a Gymnosperm. (8)

   (c) Compare the habit of Cycadales and Coniferales. (4)

6. (a) Write an account on mariculture. (6)

   (b) What are the general characters of Fungi? (7)

   (c) Give a brief account on reproduction of Lichens. (7)

7. (a) Compare the structure of thallus in *Marchantia* and *Anthoceros*. (8)

   (b) Characterize the group Coniferales. (7)

   (c) Explain the term of fossils. What are their importance? (5)

8. (a) Write about the life cycle pattern of a Rhodophyceae member. (6)

   (b) Outline the study of fossil Gymnosperms. (8)

   (c) Describe the fossil plant *Calamites*. (6)

DE–6398 **12**

DISTANCE EDUCATION

M.Sc. (Botany) (Specialization Plant Biotechnology)
DEGREE EXAMINATION, MAY 2013.

PLANT TAXONOMY

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

($5 \times 20 = 100$)

1. (a) Write an essay on structural and biological systematics. (10)

   (b) What are biotypes? Explain. (5)

   (c) Comment on phylogenetic relationship. (5)

2. (a) Explain in detail about modern system of taxonomy. (10)

   (b) Give an account on Takhtajan and Hutchinson classification. (10)

3. (a) Enumerate the procedure of Numerical taxonomy. (8)

   (b) Write the salient features of phytogeography. (6)

   (c) Write short notes on ICBN. (6)

4. (a) Comment on Herbarium. (4)

   (b) What are the criteria must follow for botanical nomenclature. Explain in detail. (10)

   (c) Give a short notes on conservation of names of species. (6)

5. (a) Discuss about diagnostic features of family *Rhamnaceae*. (10)

   (b) Write salient features of family *Orchidaceae*. (10)

6. (a) Describe biosystematics. (8)

   (b) Distinguish between Engler classification and Bentham classification. (6)

   (c) Name few important plants from the *Ranunculaceae* family. (6)

7. (a) What is the historical background of plant classification? (10)

   (b) Describe molecular systematics. (10)

8. (a) Give an account on Bentham and Hooker plant classification. (12)

   (b) Define eco species. (4)

   (c) Mention the types of species concept. (4)

DE–6399 **13**

DISTANCE EDUCATION

M.Sc. (Botany) (Specialization in Plant Biotechnology)
DEGREE EXAMINATION, MAY 2013.

ANATOMY AND EMBRYOLOGY

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carry 20 marks.

($5 \times 20 = 100$)

1. (a) Characterise the meristematic tissues. (5)

   (b) Elucidate the structure of cell wall. (10)

   (c) Describe the significance of vascular cambium. (5)

2. (a) Explain about root-stem transition. (5)

   (b) Brief on the development of vegetative organs. (10)

   (c) Write down the importance of floral vasculature. (5)

3. (a) Write down the physical, chemical and mechanical properties of wood. (8)

   (b) Briefly explain any two natural defects of wood. (7)

   (c) Write down the commercial woods of South India. (5)

4. (a) Write a detailed account on pollen morphology. (8)

   (b) Give a detailed account on megasporogenesis. (7)

   (c) Describe the structure of female gametophyte. (5)

5. (a) Write short note on polyembryony. (5)

(b) What is meant by apospory? (7)

(c) Write about the apomixis in plant improvement. (8)

6. (a) Give a general account on theories about meristem. (10)

(b) Write about vasculature in primary plant body. (5)

(c) Brief on the molecular aspects of wood differentiation. (5)

7. (a) What are the events of pollen-pistil interactions? (7)

(b) Classify the types of endosperms. (8)

(c) What is the basis of vegetative reproduction? (5)

8. (a) Write a detailed account on molecular aspects of higher plant reproduction. (7)

(b) Write an account on transfer cells. (7)

(c) Write short note on Agamospermy. (6)

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DE–6400 **14**

DISTANCE EDUCATION

M.Sc. (Botany) (Specialization – Plant Biotechnology) DEGREE EXAMINATION, MAY 2013.

PLANT TISSUE CULTURE

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

($5 \times 20 = 100$)

1. (a) Explain various types of sterilization process. (5)

   (b) Write about the function of basic elements of the tissue culture medium. (7)

   (c) Short notes on :

      (i) HEPA filter (4)

      (ii) Vegetative propagation (4)

2. (a) Mention any four surface sterilizing agent name and write advantages, applications of plant tissue culture. (8)

   (b) Write short notes on : Auxins (5)

   (c) Describe about factors affecting somatic embryogenesis. (7)

3. Write brief notes on :

   (a) Batch culture (5)

   (b) Cytokinins and Gibberellic acid (8)

   (c) Continuous culture (7)

4. (a) Write about advantages of artificial seeds. (6)

(b) Explain : Somoclonal variations (7)

(c) Explain : Protoplast culture (7)

5. (a) Discuss about the applications of micropropagation. (8)

(b) Mention any five historical events in plant tissue culture. (5)

(c) Short notes on : Significance of haploid culture. (7)

6. (a) Explain the method of somatic embryogenesis. (8)

(b) Write about the applications of secondary metabolites. (5)

(c) Short notes on : media preparation for plant tissue culture. (7)

7. (a) Write about the protoplast regeneration. (6)

(b) Explain : Somatic hybridization. (7)

(c) Describe Gynogenesis. (7)

8. (a) Write about the application somoclonal variants in plant improvement. (7)

(b) Explain : How do you produce disease free plant by tissue culture. (6)

(c) Discuss the uses of haploid in plant breeding. (7)

DE–6401 **15**

DISTANCE EDUCATION

M.Sc. (Botany) (Specialization Plant Biotechnology) DEGREE EXAMINATION, MAY 2013.

PLANT MOLECULAR BIOLOGY

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

($5 \times 20 = 100$)

1. (a) Draw and explain the Mitochondira Structure. (5)

   (b) Discuss about Targeting Gene to Chloroplast. (8)

   (c) Discuss about Plant Genome Organization. (7)

2. (a) Explain Cytoplasmic Male Sterility with an example. (8)

   (b) Describe the Classification and function seed storage protein. (10)

   (c) Define Plant Hormone and explain IDA. (2)

3. (a) Draw and explain about T DNA structure. (6)

   (b) Describe about Co integrate Vector. (6)

   (c) Give detailed account on Symbiotic nitrogen fixation with and example. (8)

4. (a) Write about on Biolistic Gene transfer. (10)

   (b) How will plant response to Stress — Salinity, UV and Osmotic pressure. (10)

5. Give a brief account on

(a) Cloning of plant Gene. (5)

(b) Explain gene Tagging. (5)

(c) Discuss about transgenic plant with two examples. (10)

6. (a) Give account on what is Chromosome, locates, structure and function. (10)

(b) Discuss about Nucleus. (10)

7. (a) Explain in detailed about Globulins and Prolamins. (6+6)

(b) Give account on Ti-Plasmid, function of Vir-genes. (8)

8. (a) Give account on plant Hormone. (8)

(b) Give account on reporter genes. (8)

(c) Discuss about nitrogen fixation Genes and its function. (4)

DE–6402 **21**

DISTANCE EDUCATION

M.Sc.(Botany) (Specialization Plant Biotechnology) DEGREE EXAMINATION, MAY 2013

PLANT PHYSIOLOGY AND BIOCHEMISTRY

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

($5 \times 20 = 100$)

1. (a) Write about the types of transpiration. (7)

   (b) Explain :

      (i) Water potential. (4)

      (ii) Chemical potential. (4)

   (c) Write short notes on: Facilitated diffusion. (5)

2. (a) How do plants absorb water? (6)

   (b) Define :

      (i) Symport. (3)

      (ii) Antiport. (3)

      (iii) Uniport and Translocation. (3)

   (c) Write about the factors affecting the photosynthesis. (5)

3. (a) Describe the oxidative phosphorylation and substrate level phosphorylation. (8)

(b) C4 carbon cycle. (6)

(c) What is the role of chloroplast pigments? (6)

4. (a) Write the process of citric acid cycle and total ATP production. (7)

(b) Write short notes on Pentose phosphate pathway. (7)

(c) Draw the structure of ATP and chloroplast with parts. (6)

5. (a) Write about the β-oxidation of fatty acids. (6)

(b) Write short notes on:

(i) Oxidoreductases. (3)

(ii) Transferases. (3)

(iii) Hydralases. (3)

(c) What are Phospholipids? Write its classification and biological functions. (5)

6. (a) Write short notes on :

(i) Disaccharides. (2)

(ii) Polysaccharides. (2)

(iii) F_1 and F_0 particles of mitochondria. (3)

(b) Give a short notes on Pyrimidine synthesis. (5)

(c) Write about the protein properties and classifications. (8)

7. (a) Write short notes on assimilation of ammonia. (5)

(b) Describe about the water transport through Tracheids and Xylem. (10)

(c) What is the mechanism of stomatal movement? (5)

8. (a) Describe about Glycolysis and explain the mechanism. (8)

(b) Explain : Symbiotic nitrogen fixation. (7)

(c) Write short notes on : Nif genes. (5).

DE–6403 **22**

DISTANCE EDUCATION

M.Sc. (Botany) (Specialization – Plant Biotechnology) DEGREE EXAMINATION, MAY 2013.

CELL BIOLOGY, GENETICS AND PLANT BREEDING

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

($5 \times 20 = 100$)

1. (a) Describe the structure and function of the Golgi complex. (6)

   (b) Describe the structure and function of chloroplast. (6)

   (c) Describe the prokaryotic cell. (8)

2. (a) Describe the processes by which ions and molecules are transported across biological membranes. (6)

   (b) What are membrane proteins, describe the various types? (7)

   (c) Describe the events taking place during prophase I of meiosis. (7)

3. (a) Comment on protein sorting in mitochondria. (6)

   (b) Describe the plasma membrane. (7)

   (c) Describe the structure and functions of lysosomes. (7)

4. (a) What is polyploidy? Describe the various types. (7)

   (b) Write short notes on penetrance and expressivity in genes. (6)

   (c) Explain maternal inheritance giving a suitable example. (7)

5. (a) Define Mendel's law of segregation and give a suitable example, to explain the law. (7)

   (b) Describe the chromosome theory of inheritance. (7)

   (c) Using a suitable example, explain extra nuclear inheritance by cellular organelles. (6)

6. (a) What is genetic variability? Elaborate its role in plant breeding. (7)

   (b) Describe a method by which a plant may be breed for stress tolerance. (6)

   (c) Write short notes on inbreeding depression theories. (7)

7. (a) Give an account of the expected mutations, as a result of non-ionizing radiations. (6)

   (b) Write short notes on gene pool and gene frequency. (7)

   (c) What are the methods by which polyploidy may be induced in plants? (7)

8. (a) What are Morgan's views on linkage? (7)

   (b) Write a note on the process of synopsis during meiosis. (7)

   (c) Explain test cross. (6)

DE–6404 **23**

DISTANCE EDUCATION

M.Sc. (Botany) (Specialisation – Plant Biotechnology) DEGREE EXAMINATION, MAY 2013.

BIOTECHNIQUES IN BOTANY

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

($5 \times 20 = 100$)

1. (a) Give short note on isoelectric focusing. (5)

   (b) Explain the concept of electrophoresis. (7)

   (c) Elaborate agarose gel electrophoresis. (8)

2. (a) Write down the principles of electron microscope. (5)

   (b) Give an account on TEM. (7)

   (c) Describe about GC. (8)

3. (a) Why radioactive compounds are unstable? Give reasons. (5)

   (b) Illustrate the applications of radioactive isotopes. (7)

   (c) Describe liquid scintillation spectrometry. (8)

4. (a) Write down the principles of centrifuge. (5)

   (b) Give an account on confocal microscopy. (7)

   (c) Describe about HPLC. (8)

5. (a) Write down the principles of blotting techniques. (5)

   (b) Explain RAPD. (7)

   (c) Describe the applications of DNA sequencing. (8)

6. (a) Give an account on autoradiography. (5)

   (b) Explain Northern Blotting techniques. (7)

   (c) Describe RFLP. (8)

7. (a) Brief any two plant genome analysis tools. (5)

   (b) Explain the relationship between genomes and proteomes. (7)

   (c) Describe tracking gene expression in plant cells. (8)

8. (a) Brief Western blotting techniques. (5)

   (b) Explain finger printing technology. (7)

   (c) What is the ultimate product of gene expression? Describe. (8)

DE–6405 **24**

DISTANCE EDUCATION

M.Sc. (Botany) (Specialization Plant Biotechnology) DEGREE EXAMINATION, MAY 2013.

BIODIVERSITY CONSERVATION AND IPR

(2008 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

($5 \times 20 = 100$)

1. (a) Write notes on Agrobiodiversity. (6)

   (b) Discuss Biodiversity act of India 2002 and 2004. (9)

   (c) Mention the types of biodiversity. (5)

2. (a) Explain the vegetation types of India and TamilNadu. (10)

   (b) Write on patenting life forms and their impact on biodiversity. (6)

   (c) Write short notes on Red Data Book. (4)

3. (a) Give on detail account of sustainable use of plant genetic resources. (8)

   (b) Write an essay about *in situ* conservation. (10)

   (c) Role of germplasm storage. (2)

4. (a) Describe about the establishment and functions of GATT. (8)

   (b) Write the significance of intellectual property rights. (6)

   (c) Short notes on plant breeders rights. (6)

5. (a) Discuss the advantages and disadvantages of GM foods. (5)

   (b) Describe General guidelines for research in transgenic plants. (7)

   (c) Write notes on GLP. (8)

6. (a) Explain in detail about values and uses of biodiversity. (6)

   (b) Write an essay about National parks and Biosphere reserves. (10)

   (c) What do you think about irregularities of patenting? (4)

7. (a) Explain loss of biodiversity. (6)

   (b) Describe general overview of plant conservation. (10)

   (c) Hotspot biodiversity areas in India. (4)

8. (a) Explain about WTO. (7)

   (b) Distinguish between genetically modified and conventional plant resources. (6)

   (c) Describe centers of biodiversity. (7)

DE–6406 **25**

DISTANCE EDUCATION

M.Sc.(Botany) (Specialization Plant Biotechnology)
DEGREE EXAMINATION, MAY 2013.

PLANT GENETIC ENGINEERING

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

($5 \times 20 = 100$)

1. (a) Enumerate the significance of genetically engineered plants. (8)

   (b) Give the detailed account on Restriction and polymerase enzymes. (8)

   (c) How would you correlate modern cloning with Gregory Mendel? (4)

2. (a) Write short notes on Golden Rice. (4)

   (b) Brief about selectable markers and their role in plant transformation. (10)

   (c) Explain the carrier molecule in plant genetic engineering. (6)

3. (a) Give detailed protocol of Biolistic method. (5)

   (b) Discuss the role of Ti plasmid in the development of disease resistant crops. (10)

   (c) List out any two novel transformation techniques and their mechanism in plant cell. (5)

4. Define briefly:

(a) CMS (7)

(b) Thermo cycler. (7)

(c) Gene Bank and cryopreservation. (6)

5. (a) How antibiotic markers beneficial in cloning. (5)

(b) Describe the eukaryotic photosynthetic organelle engineering. (10)

(c) Write the significance of molecular marker STS. (5)

6. (a) Essay about Promoters used in plant. (7)

(b) Derive the action mechanism of Proteinase inhibitor on insects. (8)

(c) Write short notes on Gene silencing. (5)

7. (a) Describe the manipulation of Abiotic stress tolerance plants. (7)

(b) Write an essay on about 'Golden Rice Project'. (10)

(c) Protoplast fusion. (3)

8. (a) Write short notes on SCAR. (5)

(b) Mention the tools of plant genetic engineering. (10)

(c) Explain – Flavr Savr. (5)